

Lesson 1 Position Calibration

Note: Please calibrate the angle and position of camera before going to the programming games. Otherwise, it will affect the recognition range, and then affect the implementation of the games.

1. Getting Ready

1.1 APP Installation

Android User: Please send the installation pack in the folder to smart phone to install.

iOS User: Please search “WonderPi” in the APP Store.

Don't forget to open the GPS location and Wi-Fi service before going to the APP.

1.2 Turn on ArmPi

Place and ArmPi on the map. Connect and switch on Raspberry Pi expansion board. The LED1 and LED2 of the Raspberry Pi will be on firstly and then the LED2 will flash every 2 seconds.

2. Connect ArmPi

After turning on successfully, Raspberry Pi will launch a Wi-Fi hotspot named with the first letters “HW”. Please choose to connect it.

1) Open the APP, choose "ArmPi" and then click the "+" button to choose "Direct Connection Mode".



2) Go to Settings in the smart phone to connect the hotspot with the first letters "HW".



Connect your phone/iPad to Device hotspot:
"HW-xxxxxxx", and return to app.

[Go to connect device hotspots](#)

< Select WLAN

WLAN



CONNECTED



HW-4ABE9C8F

Connected, no internet



3) After connecting, open the APP. You can find that device connect with the APP automatically. Wait until the robotic arm icon shown in the interface, which means the connection is successful.



i If the pop-up window shows "The network is unavailable, continue to connect or not", click "Keep connected".

There are two methods "Direct Connection Mode" and "LAN Mode" to connect the Raspberry Pi hotspot.

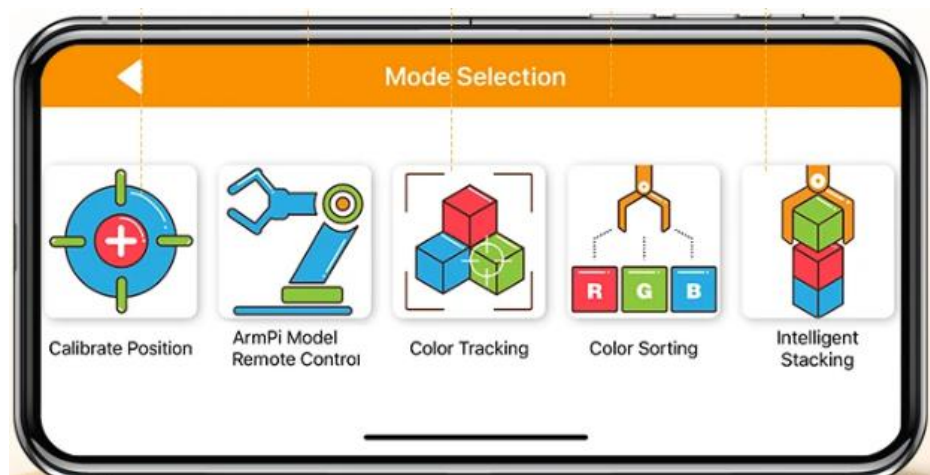
3. Position Calibration

3.1 Calibration Standard

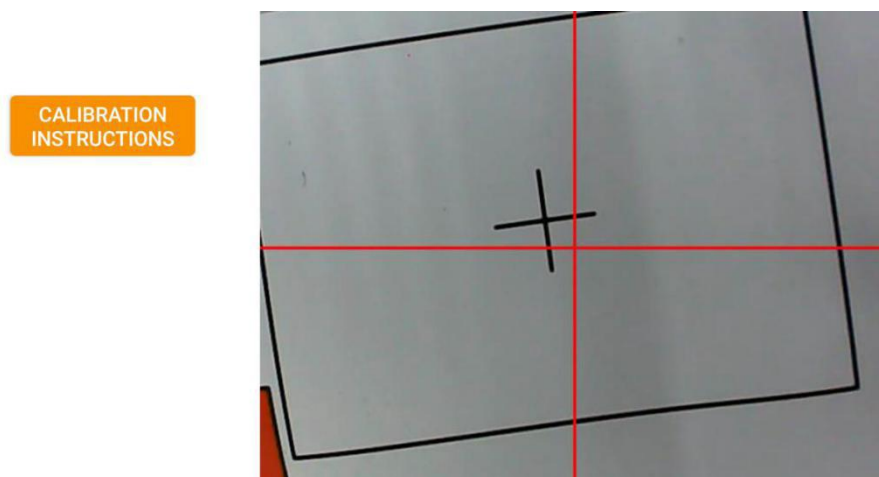
- 1) Rotate the camera to the map.
- 2) Calibrate the red cross on the camera screen to completely overlap the black cross on the map.

3.2 Calibrate Steps

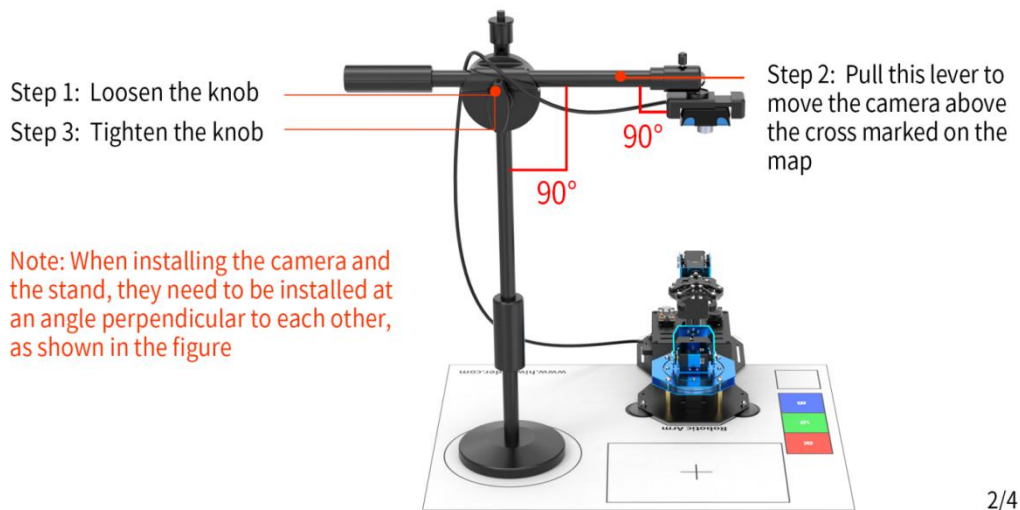
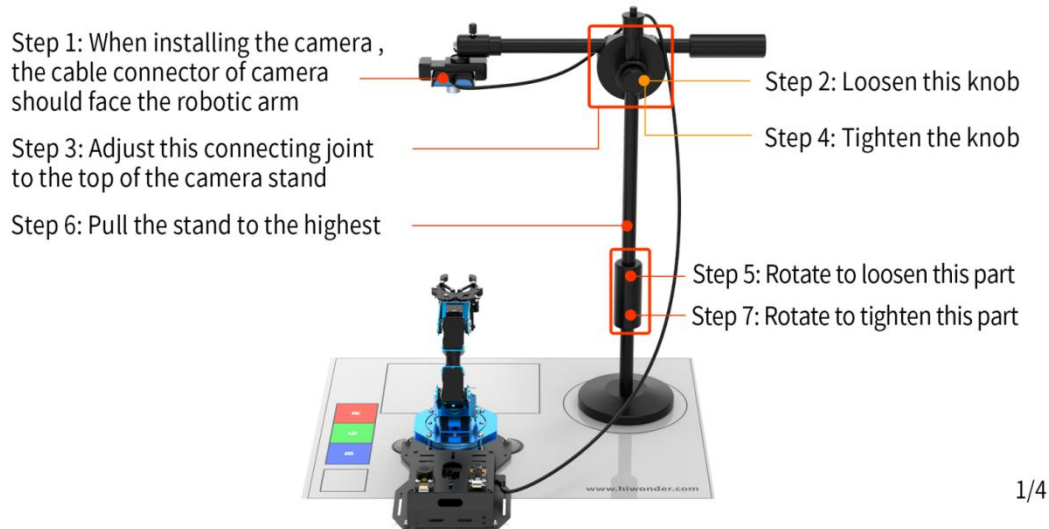
- 1) Click the robotic arm icon to enter the “Mode Selection”.



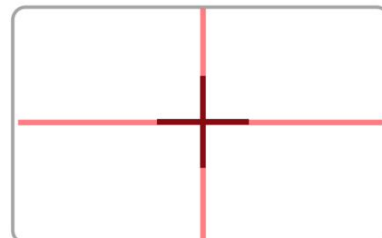
- 2) Click “Calibrate Position”.



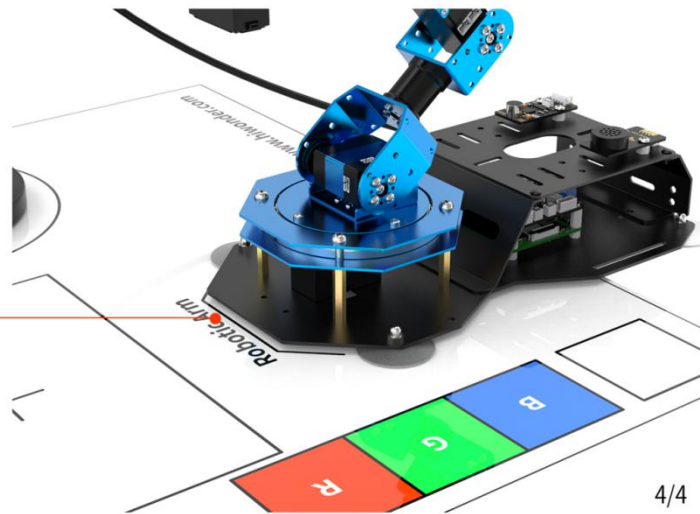
3) Click "Calibration instruction".



Move the base of the camera stand to adjust the camera's field of view. In this step, the red cross on the camera screen needs to completely overlap the black cross on the map



Place the suction cups on the front of the robotic arm in the hollow circle on the map, and then align the bottom of the robotic arm with the line in the figure



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